

## ★ 1. ELECTRIC CHARGES & PROPERTIES

### ✓ Electric Charge

A basic property of matter that causes electric forces.

### ✓ Types

- Positive charge
- Negative charge

Like charges repel; unlike charges attract.

### ✓ Properties

#### 1. Additivity:

$$Q_{\text{total}} = Q_1 + Q_2 + Q_3 + \dots$$

#### 2. Quantization:

$$Q = ne, \quad e = 1.6 \times 10^{-19} \text{ C}$$

#### 3. Conservation:

Charge can neither be created nor destroyed.

### ✓ Methods of Charging

- Friction (rubbing)
- Conduction (contact)
- Induction (no contact, very important)

## ★ 2. CONDUCTORS, INSULATORS & INDUCTION

### ✓ Conductors

Have free electrons

Examples: Cu, Al, metals

### ✓ Insulators

No free electrons

Examples: Rubber, glass

### ✓ Induction (Key Concept)

Bring a negative rod near a metal sphere:

(- - - - -) | + - |

Ground the far end → negative charges escape → sphere becomes **positively charged**.

### ✓ Electrostatic Shielding

Inside a closed conductor,

$$E = 0$$

### ★ 3. COULOMB'S LAW

Force between two point charges:

$$F = k \frac{q_1 q_2}{r^2}$$

Where  $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$ .

#### ✓ Vector Form

$$\vec{F}_{12} = k \frac{q_1 q_2}{r^2} \hat{r}$$

#### ✓ Superposition

Total force is vector sum of individual forces.

### ★ 4. ELECTRIC FIELD (E)

Electric field at a point:

$$\vec{E} = \frac{\vec{F}}{q_0}$$

#### ✓ Electric field due to a point charge

$$E = k \frac{Q}{r^2}$$

#### ✓ Direction

Outward from + charge, inward towards – charge.

#### ✓ Field Lines

- Never intersect
- Start on +, end on –
- Density  $\propto$  field strength

### ★ 5. ELECTRIC DIPOLE

Dipole = +q and –q separated by distance 2a.

✓ Dipole Moment

$$\vec{p} = q \cdot 2a$$

✓ Field on axial line

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$$

✓ Field on equatorial line

$$E_{\text{eq}} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3}$$

✓ Torque in uniform E-field

$$\tau = pE \sin \theta$$

## ★ 6. ELECTRIC FLUX ( $\Phi$ )

$$\phi = \vec{E} \cdot \vec{A} = EA \cos \theta$$

Unit:  $\text{N} \cdot \text{m}^2/\text{C}$

Flux represents the number of field lines passing through a surface.

## ★ 7. GAUSS'S LAW

$$\phi = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

✓ Applications

(a) Infinite Line Charge

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

(b) Infinite Plane Sheet

$$E = \frac{\sigma}{2\epsilon_0}$$

(c) Spherical Shell

Inside:

$$E = 0$$

Outside:

$$E = k \frac{Q}{r^2}$$

#### (d) Solid Sphere

If  $r < R$ :

$$E = \frac{Qr}{4\pi\epsilon_0 R^3}$$

If  $r \geq R$ : same as point charge.

### ★ 8. ELECTRIC POTENTIAL (V)

Work done per unit positive charge.

$$V = \frac{W}{q}$$

#### ✓ Potential due to a point charge

$$V = k \frac{Q}{r}$$

#### ✓ Potential due to dipole

$$V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$$

### ★ 9. EQUIPOTENTIAL SURFACES

Surfaces having same potential everywhere.

#### ✓ Properties

- No work is done along an equipotential surface
- Electric field is **perpendicular** to equipotential surface
- Equipotentials closer → stronger E-field

### ★ 10. ELECTRIC POTENTIAL ENERGY

#### ✓ For two point charges

$$U = k \frac{q_1 q_2}{r}$$

✓ Energy of dipole in uniform field

$$U = -pE \cos \theta$$

## ★ 11. CAPACITANCE

$$C = \frac{Q}{V}$$

Unit: Farad (F)

✓ Parallel Plate Capacitor

$$C = \frac{\epsilon_0 A}{d}$$

Dependence:

- $\uparrow$  Area  $\rightarrow \uparrow C$
- $\uparrow$  Distance  $\rightarrow \downarrow C$

## ★ 12. COMBINATION OF CAPACITORS

✓ Series

$$\frac{1}{C_{\text{eq}}} = \sum \frac{1}{C_i}$$

✓ Parallel

$$C_{\text{eq}} = C_1 + C_2 + C_3 + \dots$$

## ★ 13. ENERGY STORED IN CAPACITOR

$$U = \frac{1}{2} CV^2 = \frac{Q^2}{2C} = \frac{1}{2} QV$$

✓ Energy Density

$$u = \frac{1}{2} \epsilon_0 E^2$$

## ★ 14. DIELECTRICS & POLARIZATION

### ✓ Dielectric

An insulating material placed between plates.

### ✓ Dielectric Constant (K)

$$C = KC_0$$

### ✓ Capacitor with dielectric slab

$$C = \frac{K\epsilon_0 A}{d}$$

### ✓ Polarization

Alignment of molecules when dielectric is placed in electric field → increases capacitance.

## 🎁 CHAPTER SUMMARY (SUPER QUICK)

- Charges exert forces → Coulomb's law
- Electric field → force per unit charge
- Gauss's law → powerful tool for symmetric charge distributions
- Potential → work per unit charge
- Capacitors store energy
- Dielectrics increase capacitance